

Cloud Computing Assignment (Solved)

Q1. Key Characteristics of Cloud Computing and Difference from Traditional Computing

Cloud computing is a technology that provides computing resources such as servers, storage, and applications over the internet. Its key characteristics include on-demand self-service, where users can access resources without human interaction; broad network access, which allows services to be accessed from anywhere; resource pooling, where multiple users share the same infrastructure; rapid scalability, which enables quick increase or decrease of resources; and measured service, where users pay only for what they use. Unlike traditional computing, which requires physical hardware and high upfront costs, cloud computing is flexible, cost-efficient, and managed by service providers.

Q2. Cloud Service Models (IaaS, PaaS, SaaS) with Examples

Cloud computing offers three main service models. Infrastructure as a Service (IaaS) provides virtualized hardware resources such as servers and storage; examples include Amazon EC2 and Microsoft Azure Virtual Machines. Platform as a Service (PaaS) offers a platform for developers to build and deploy applications without managing infrastructure; examples include Google App Engine and Heroku. Software as a Service (SaaS) delivers ready-to-use applications over the internet, such as Google Docs, Gmail, and Microsoft Office 365. Each model offers different levels of control and responsibility for users.

Q3. Scalability, Flexibility, and Cost Benefits of Cloud Computing

Cloud computing enhances scalability by allowing businesses to easily scale resources up or down based on demand. This flexibility helps organizations respond quickly to changing workloads without investing in additional hardware. Cloud services reduce costs by eliminating the need for expensive infrastructure, maintenance, and IT staff. The pay-as-you-go model ensures that businesses only pay for the resources they use, making cloud computing a cost-effective solution, especially for startups and small businesses.

Q4. Data Security and Privacy in Cloud Computing

Data security and privacy are critical concerns in cloud computing because sensitive information is stored online. To ensure protection, cloud providers implement measures such as data encryption, secure authentication, firewalls, regular security updates, and compliance with international security standards. Organizations can also use access control policies, multi-factor authentication, and regular backups to protect their data. By following these practices, cloud computing can provide a secure environment for storing and managing information.